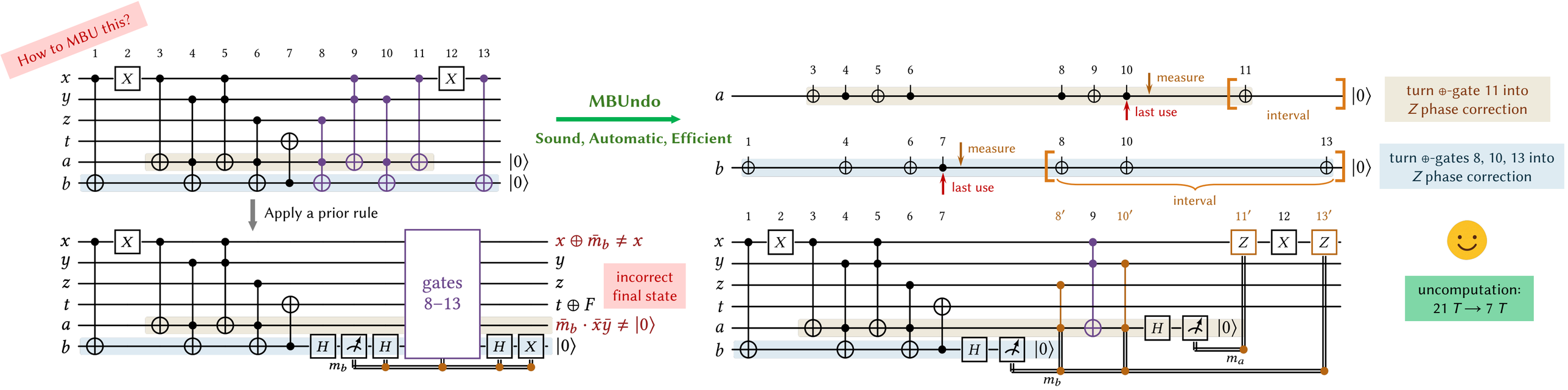

How to MBU this?
1 2 3 4 5 6 7 8 9 10 11 12 13
x y z t a b
X X
|0⟩ |0⟩
Apply a prior rule
gates 8–13
H H H X
$m_b$
$x \oplus \bar{m}_b \neq x$
y
z
$t \oplus F$
$\bar{m}_b \cdot \bar{x}\bar{y} \neq |0\rangle$
|0⟩
incorrect final state
MBUndo
Sound, Automatic, Efficient
a
3 4 5 6 8 9 10 measure 11
last use
interval
|0⟩
b
1 4 6 7 measure 8 10 13
last use
interval
|0⟩
turn ⊕-gate 11 into Z phase correction
turn ⊕-gates 8, 10, 13 into Z phase correction
1 2 3 4 5 6 7 8′ 9 10′ 11′ 12 13′
x y z t a b
X H |0⟩ H |0⟩ Z X Z
$m_b$ $m_a$
uncomputation: 21 T → 7 T